



HPC

User environment

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User workspace

The default login shell for your user account is `bash`.

To determine your current login shell:

```
$ echo $SHELL
```

By default, the login shell change directory to `$HOME` folder.

Each user has 2 folders (`$HOME` and `$WORK`) organized as follows:

Directory	Environment variable	File system	Backup	Quota
<code>/Home/Users/<login></code>	<code>\$HOME</code>	NFS	Yes	10 GB
<code>/Work/Users/<login></code>	<code>\$WORK</code>	BeeGFS	No	1 TB
<code>/Work/Projects/<name></code>		BeeGFS	No	3 TB

! DIRECTORIES

- `$WORK` is used as working directory: all simulation data must be copied in this folder.
- `$HOME` is read-only in computing nodes.

Using modules to manage your environment

Installed applications are organized using `module` command according to the architecture of the node (Intel, AMD).

The `module` command supports the following subcommands:

Command	Description
<code>module display <module></code>	Display what a module does
<code>module list</code>	List loaded modules
<code>module avail</code>	List all the modules which are available to be loaded
<code>module load <mod1> [mod2...]</code>	Load a module
<code>module unload <module></code>	Unload a module
<code>module purge</code>	Unload all modules (purge)

For example on the login node, one can view **all** installed/available package software:

```
$ module avail -t
/Softs/helios-modules:
ansys/2021R1
ansys/2022R2
apptainer/1.1.8
deep/pytorch-gpu/2.0.1
deep/rapids-gpu/23.04
deep/tensorflow-gpu/2.12.0
energy_scope/2022-05-05
gurobi/10.0.2
matlab/r2021b
matlab/r2023a
moose/gcc-12.1.0
nix/2.11.0
openradioass/20231106
tools/bedtools/2.29.1
tools/turbovnc/local
travis/20210521
```

```
/Softs/spack/share/spack/modules/linux-rocky8-zen2:  
anaconda3@2021.05/gcc-12.1.0  
gcc@12.1.0/gcc-12.1.0  
r@4.1.3/gcc-12.1.0
```

```
/Softs/spack/share/spack/modules/linux-rocky8-x86_64:  
anaconda3@2022.10/gcc-12.1.0  
miniconda3@4.10.3/gcc-12.1.0  
nextflow@23.04.3/gcc-12.1.0
```

```
/Softs/spack/share/spack/modules/linux-rocky8-skylake_avx512:  
anaconda3@2021.05/gcc-12.1.0  
berkeley-db@18.1.40/gcc-12.1.0  
cp2k@8.2/openmpi-2.1.6-gcc-12.1.0-openblas  
freefem@4.9/openmpi-2.1.6-gcc-12.1.0-openblas  
gcc@11.3.0/gcc-12.1.0  
gcc@12.1.0/gcc-8.5.0  
gms@4.8.4/openmpi-2.1.6-gcc-12.1.0-openblas  
gromacs@2021.5/openmpi-2.1.6-gcc-12.1.0  
intel-oneapi-advisor@2022.0.0/gcc-12.1.0  
intel-oneapi-compilers@2022.0.2/gcc-12.1.0  
ior@3.3.0/openmpi-2.1.6-gcc-12.1.0  
namd@2.14/openmpi-2.1.6-gcc-12.1.0  
nextflow@22.04.0/gcc-12.1.0  
octave@6.4.0/openmpi-2.1.6-gcc-12.1.0-openblas  
octave@7.1.0/openmpi-2.1.6-gcc-12.1.0-openblas  
octopus@11.4/openmpi-2.1.6-gcc-11.3.0-openblas  
opa-psm2@11.2.185/gcc-12.1.0  
openjdk@11.0.14.1_1/gcc-12.1.0  
openmpi@2.1.6/gcc-12.1.0  
openmpi@2.1.6/intel-2021.5.0  
openmpi@4.1.2/gcc-12.1.0  
openmx@3.8/openmpi-2.1.6-gcc-12.1.0-openblas  
perl@5.36.0/gcc-12.1.0  
quantum-espresso@7.0/openmpi-2.1.6-gcc-12.1.0-openblas  
r@4.1.3/gcc-12.1.0  
siesta@4.0.2/openmpi-2.1.6-gcc-12.1.0-openblas  
vasp@5.4.4/openmpi-2.1.6-gcc-12.1.0-openblas  
vasp@6.2.0/openmpi-2.1.6-gcc-12.1.0-openblas
```

! MODULES

- The management of modules according to the node architecture is provided by Spack software.
- Module name is generated according to the software version, compiler version and even MPI version if enabled.

For example to load and use `namd@2.14/openmpi-2.1.6-gcc-12.1.0` (NAMD version 2.14 compiled with GCC-12.1.0 and OpenMPI 2.1.6)

```
$ module load namd@2.14/openmpi-2.1.6-gcc-12.1.0
```

Transfer data from/to Helios



TIP

For Windows users, if you transfer textual files (Slurm scripts or text input files) to Helios, please use `dos2unix` on them after copy.

Usage:

```
[user@mesointeractive ~] $ dos2unix <files...>
```

Using SSH command line

⚠ WARNING

These commands **must** be executed on your `local` machine !

- To copy files from your `localhost` to `Helios`:

```
$ scp file1 file2 login@mesohelios1.univ-fcomte.fr:WORK
```

This command will copy `file1` and `file2` to directory `WORK` on Helios.

- To copy files from **Helios** to your **localhost**:

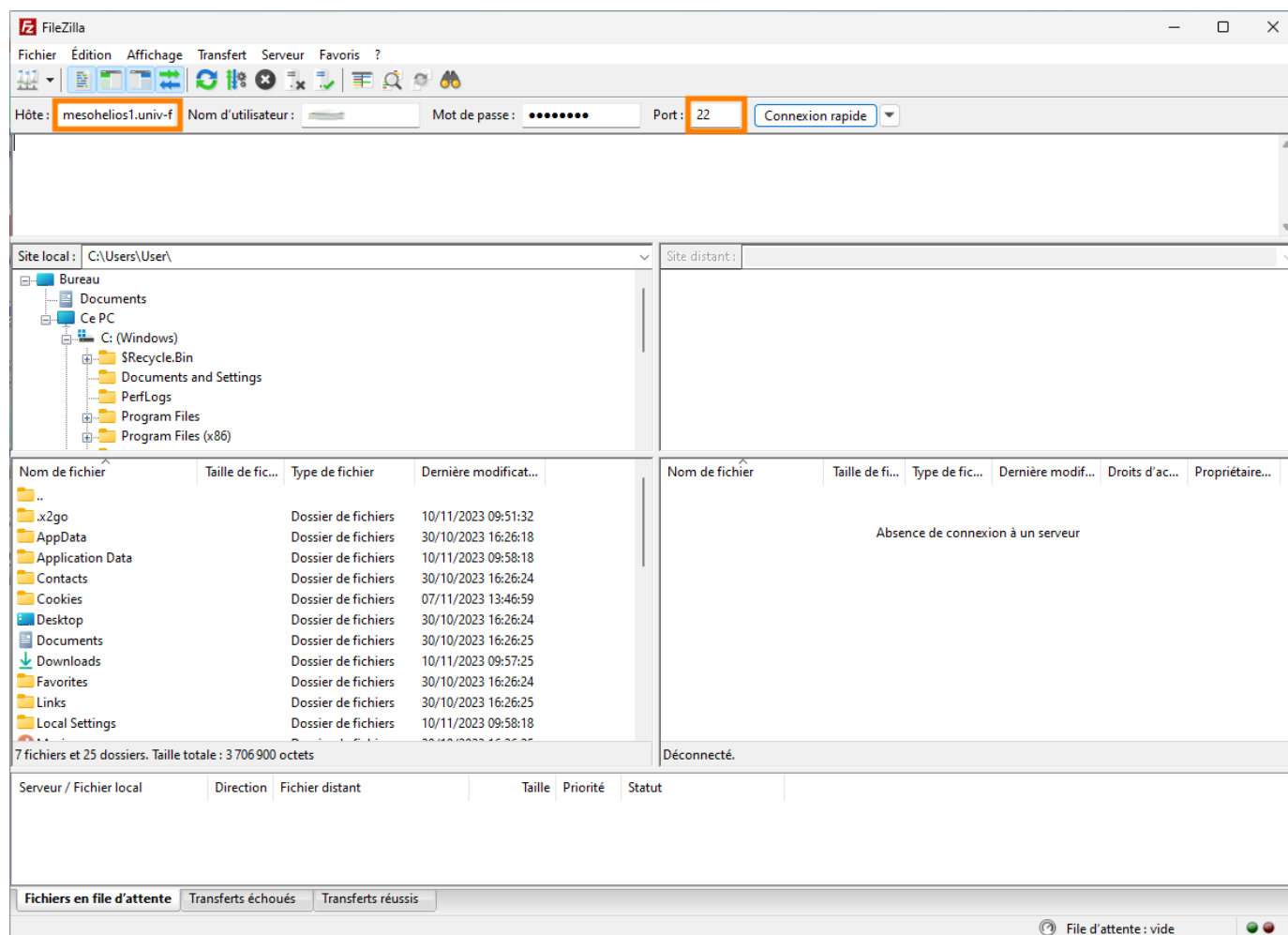
```
$ scp login@mesohelios1.univ-fcomte.fr:WORK/files.tgz .
```

This command will copy **files.tgz** from directory **WORK** on Helios to your current directory on localhost.

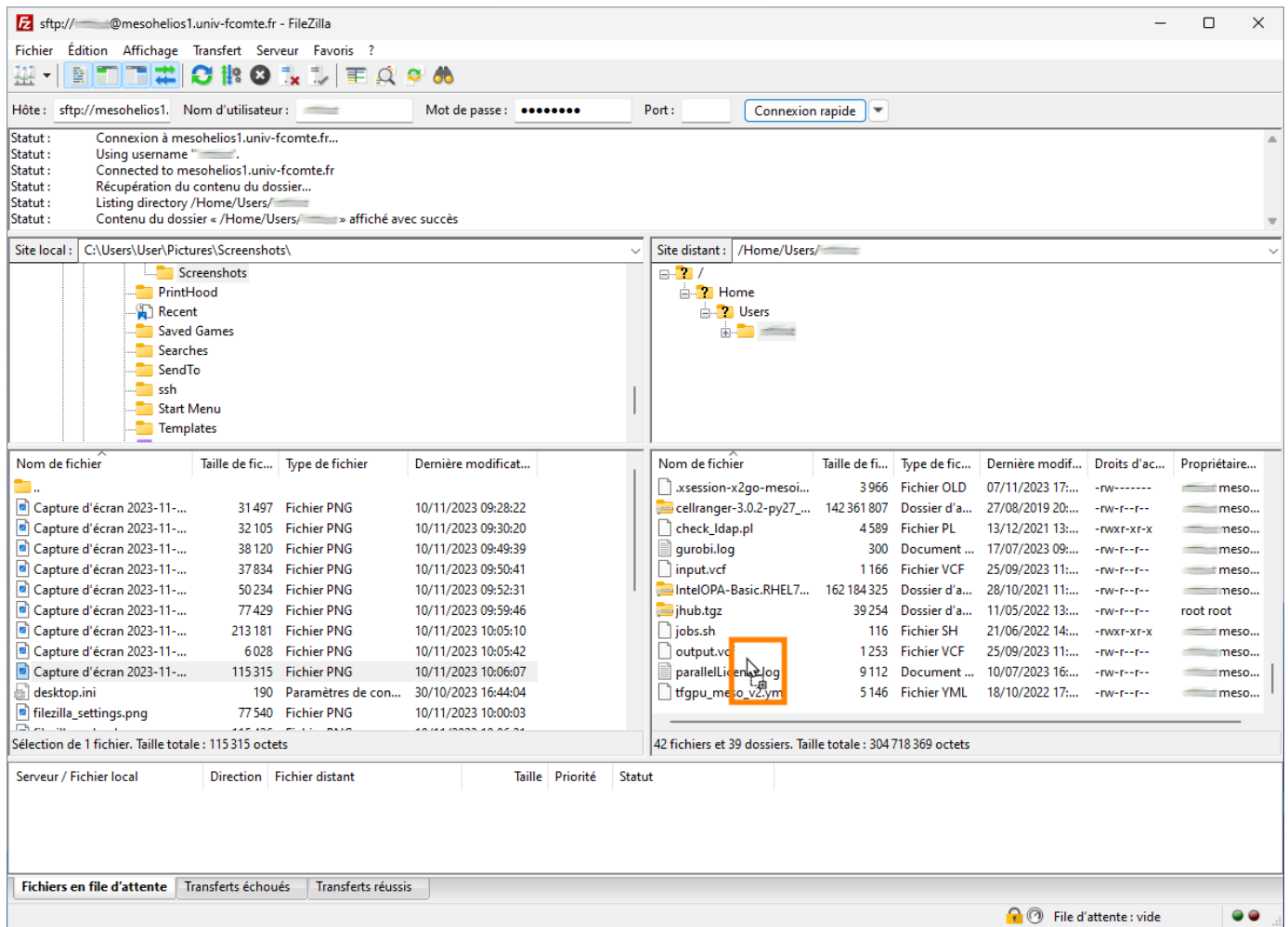
Using graphical tools

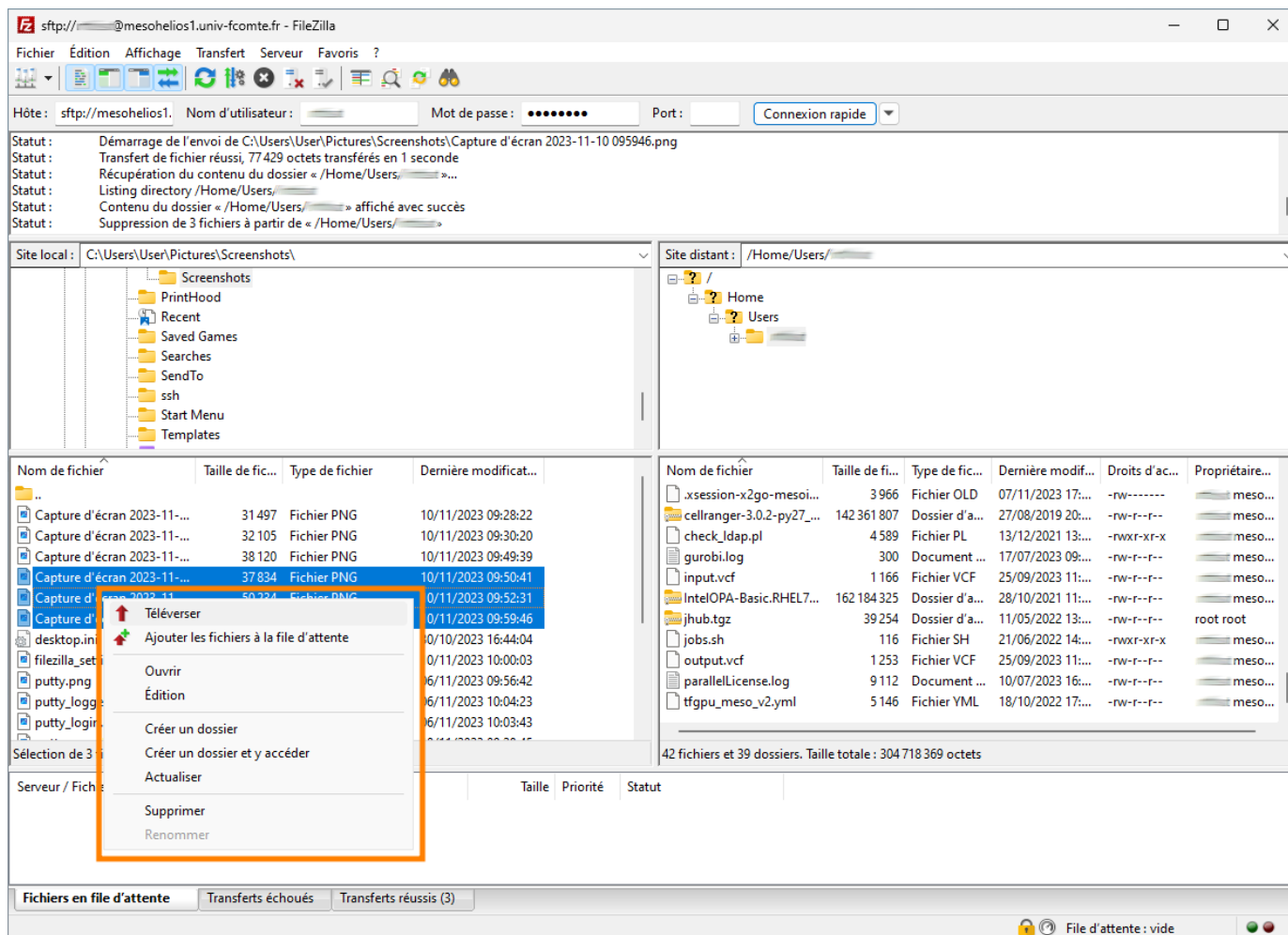
FileZilla is a good Graphical User Interface (GUI) to SFTP protocol (based on SSH).

Use the same credentials as with ssh/scp and use port 22 to use SFTP protocol.



Once connected, you can simply transfer files by drag & drop operations or by right-clicking in the file(s) you wish to transfer:





Citizenship on Helios

Be always aware that you are working on a **shared system** where your behavior could have a negative impact on the workflow of other users.